

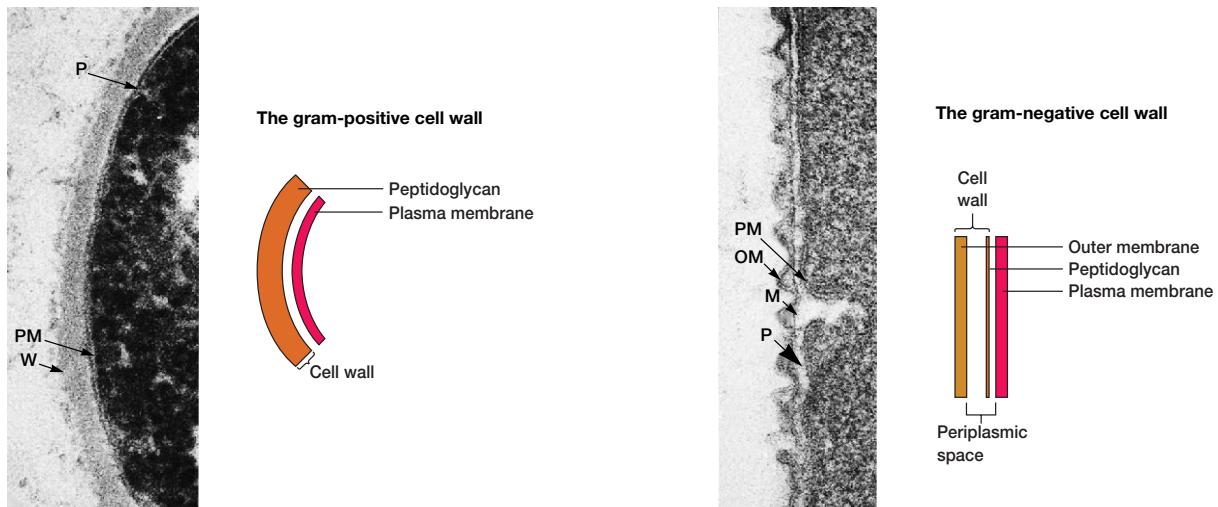


Figure 3.15 Gram-Positive and Gram-Negative Cell Walls. The gram-positive envelope is from *Bacillus licheniformis* (left), and the gram-negative micrograph is of *Aquaspirillum serpens* (right). M; peptidoglycan or murein layer; OM, outer membrane; PM, plasma membrane; P, periplasmic space; W, gram-positive peptidoglycan wall.

3.5 The Prokaryotic Cell Wall

The cell wall is the layer, usually fairly rigid, that lies just outside the plasma membrane. It is one of the most important parts of a prokaryotic cell for several reasons. Except for the mycoplasmas (see section 23.1) and some Archaea (see chapter 20), most bacteria have strong walls that give them shape and protect them from osmotic lysis (p. 61); wall shape and strength is primarily due to peptidoglycan, as we will see shortly. The cell walls of many pathogens have components that contribute to their pathogenicity. The wall can protect a cell from toxic substances and is the site of action of several antibiotics.

After Christian Gram developed the Gram stain in 1884, it soon became evident that bacteria could be divided into two major groups based on their response to the Gram-stain procedure (see table 19.9). Gram-positive bacteria stained purple, whereas gram-negative bacteria were colored pink or red by the technique. The true structural difference between these two groups became clear with the advent of the transmission electron microscope. The gram-positive cell wall consists of a single 20 to 80 nm thick homogeneous **peptidoglycan** or **murein** layer lying outside the plasma membrane (figure 3.15). In contrast, the gram-negative cell wall is quite complex. It has a 2 to 7 nm peptidoglycan layer surrounded by a 7 to 8 nm thick **outer membrane**. Because of the thicker peptidoglycan layer, the walls of gram-positive cells are stronger than those of gram-negative bacteria. Microbiologists often call all the structures from the plasma membrane outward the **envelope** or cell envelope. This includes the wall and structures like capsules (p. 61) when present. [Gram-stain procedure \(p. 28\)](#)

Frequently a space is seen between the plasma membrane and the outer membrane in electron micrographs of gram-negative bacteria, and sometimes a similar but smaller gap may

be observed between the plasma membrane and wall in gram-positive bacteria. This space is called the **periplasmic space**. Recent evidence indicates that the periplasmic space may be filled with a loose network of peptidoglycan. Possibly it is more a gel than a fluid-filled space. The substance that occupies the periplasmic space is the **periplasm**. Gram-positive cells may have periplasm even if they lack a discrete, obvious periplasmic space. Size estimates of the periplasmic space in gram-negative bacteria range from 1 nm to as great as 71 nm. Some recent studies indicate that it may constitute about 20 to 40% of the total cell volume (around 30 to 70 nm), but more research is required to establish an accurate value. When cell walls are disrupted carefully or removed without disturbing the underlying plasma membrane, periplasmic enzymes and other proteins are released and may be easily studied. The periplasmic space of gram-negative bacteria contains many proteins that participate in nutrient acquisition—for example, hydrolytic enzymes attacking nucleic acids and phosphorylated molecules, and binding proteins involved in transport of materials into the cell. Denitrifying and chemolithoautotrophic bacteria (see sections 9.6 and 9.10) often have electron transport proteins in their periplasm. The periplasmic space also contains enzymes involved in peptidoglycan synthesis and the modification of toxic compounds that could harm the cell. Gram-positive bacteria may not have a visible periplasmic space and do not appear to have as many periplasmic proteins; rather, they secrete several enzymes that ordinarily would be periplasmic in gram-negative bacteria. Such secreted enzymes are often called **exoenzymes**. Some enzymes remain in the periplasm and are attached to the plasma membrane.

The Archaea differ from other prokaryotes in many respects (see chapter 20). Although they may be either gram positive or gram negative, their cell walls are distinctive in structure and

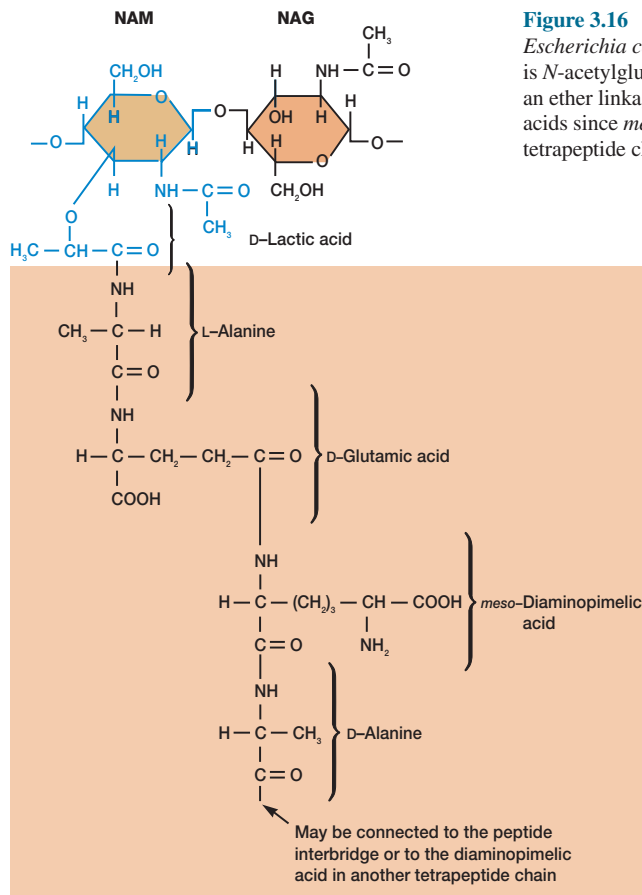


Figure 3.16 Peptidoglycan Subunit Composition. The peptidoglycan subunit of *Escherichia coli*, most other gram-negative bacteria, and many gram-positive bacteria. NAG is *N*-acetylglucosamine. NAM is *N*-acetylmuramic acid (NAG with lactic acid attached by an ether linkage). The tetrapeptide side chain is composed of alternating D- and L-amino acids since *meso*-diaminopimelic acid is connected through its L-carbon. NAM and the tetrapeptide chain attached to it are shown in different shades of color for clarity.

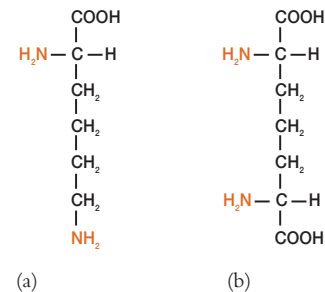


Figure 3.17 Diaminoacids Present in Peptidoglycan. (a) L-Lysine. (b) *meso*-Diaminopimelic acid.

chemical composition. The walls lack peptidoglycan and are composed of proteins, glycoproteins, or polysaccharides.

Following this overview of the envelope, peptidoglycan structure and the organization of gram-positive and gram-negative cell walls are discussed in more detail.

Peptidoglycan Structure

Peptidoglycan or murein is an enormous polymer composed of many identical subunits. The polymer contains two sugar derivatives, *N*-acetylglucosamine and *N*-acetylmuramic acid (the lactyl ether of *N*-acetylglucosamine), and several different amino acids, three of which—D-glutamic acid, D-alanine, and *meso*-diaminopimelic acid—are not found in proteins. The presence of D-amino acids protects against attack by most peptidases. The peptidoglycan subunit present in most gram-negative bacteria and many gram-positive ones is shown in **figure 3.16**. The backbone of this polymer is composed of alternating *N*-acetylglucosamine and *N*-acetylmuramic acid residues. A peptide chain of four alternating D- and L-amino acids is connected to the carboxyl group of *N*-acetylmuramic acid. Many bacteria substitute another diaminoacid, usually L-lysine, in the third position for *meso*-diaminopimelic acid (**figure 3.17**). [A review of the](#)

[chemistry of biological molecules \(appendix I\); Peptidoglycan structural variations \(pp. 521–22\)](#)

Chains of linked peptidoglycan subunits are joined by cross-links between the peptides. Often the carboxyl group of the terminal D-alanine is connected directly to the amino group of diaminopimelic acid, but a **peptide interbridge** may be used instead (**figure 3.18**). Most gram-negative cell wall peptidoglycan lacks the peptide interbridge. This cross-linking results in an enormous peptidoglycan sac that is actually one dense, interconnected network (**figure 3.19**). These sacs have been isolated from gram-positive bacteria and are strong enough to retain their shape and integrity (**figure 3.20**), yet they are elastic and somewhat stretchable, unlike cellulose. They also must be porous, as molecules can penetrate them.

Gram-Positive Cell Walls

Normally the thick, homogeneous cell wall of gram-positive bacteria is composed primarily of peptidoglycan, which often contains a peptide interbridge (**figure 3.20** and **figure 3.21**). However gram-positive cell walls usually also contain large amounts of **teichoic acids**, polymers of glycerol or ribitol joined by phosphate

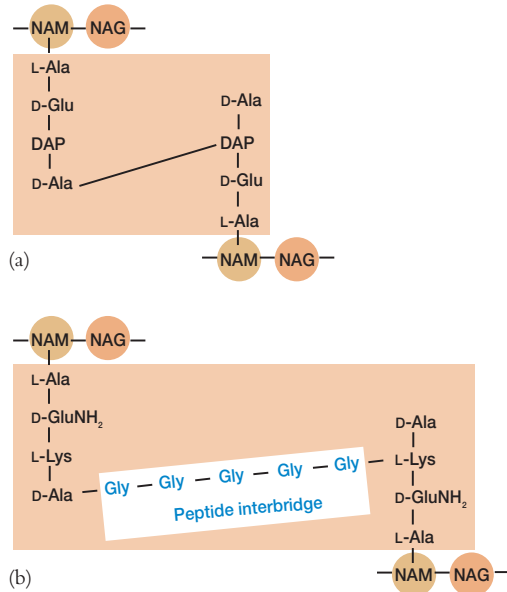


Figure 3.18 Peptidoglycan Cross-Links. (a) *E. coli* peptidoglycan with direct cross-linking, typical of many gram-negative bacteria. (b) *Staphylococcus aureus* peptidoglycan. *S. aureus* is a gram-positive bacterium. NAM is *N*-acetylmuramic acid. NAG is *N*-acetylglucosamine. Gly is glycine. Although the polysaccharide chains are drawn opposite each other for the sake of clarity, two chains lying side-by-side may be linked together (see figure 3.19).

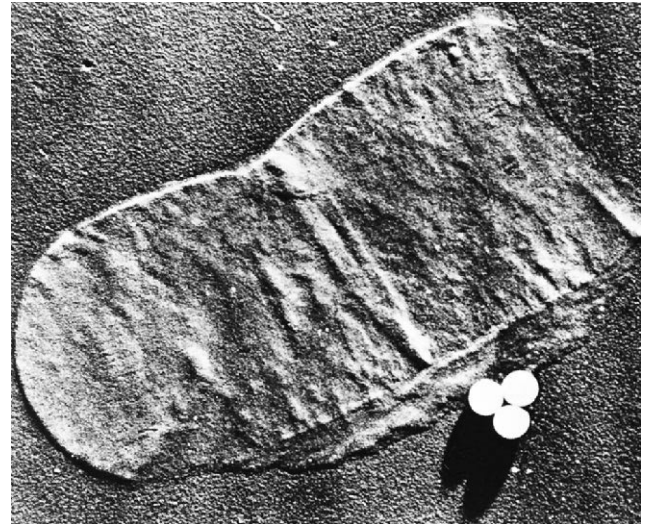


Figure 3.20 Isolated Gram-Positive Cell Wall. The peptidoglycan wall from *Bacillus megaterium*, a gram-positive bacterium. The latex spheres have a diameter of 0.25 μm.

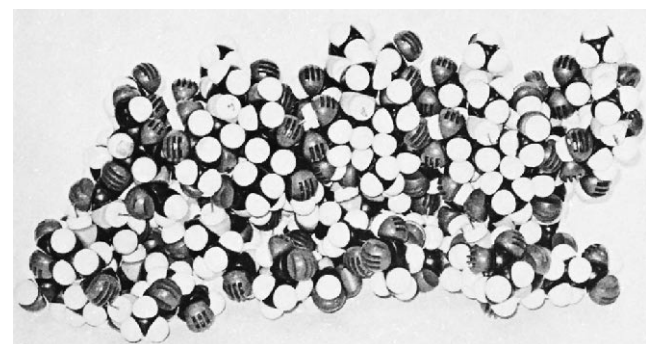
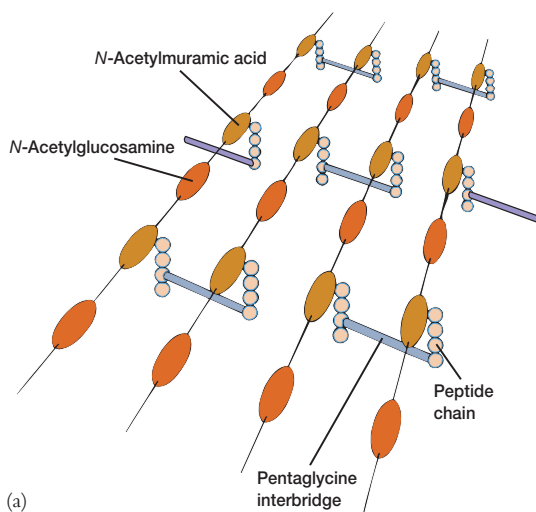


Figure 3.19 Peptidoglycan Structure. A peptidoglycan segment showing the polysaccharide chains, tetrapeptide side chains, and peptide interbridges. (a) A schematic diagram. (b) A space-filling model of gram-negative murein with four repeating peptidoglycan subunits in the plane of the paper. Two chains are arranged vertical to this direction.

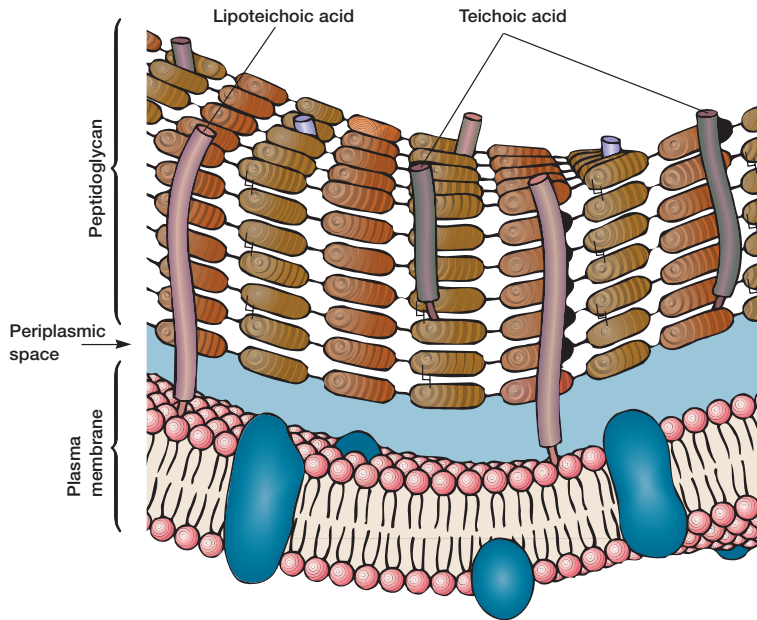


Figure 3.21 The Gram-Positive Envelope.

groups (figures 3.21 and 3.22). Amino acids such as D-alanine or sugars like glucose are attached to the glycerol and ribitol groups. The teichoic acids are connected to either the peptidoglycan itself by a covalent bond with the six hydroxyl of *N*-acetylmuramic acid or to plasma membrane lipids; in the latter case they are called lipoteichoic acids. Teichoic acids appear to extend to the surface of the peptidoglycan, and, because they are negatively charged, help give the gram-positive cell wall its negative charge. The functions of these molecules are still unclear, but they may be important in maintaining the structure of the wall. Teichoic acids are not present in gram-negative bacteria.

Gram-Negative Cell Walls

Even a brief inspection of figure 3.15 shows that gram-negative cell walls are much more complex than gram-positive walls. The thin peptidoglycan layer next to the plasma membrane may constitute not more than 5 to 10% of the wall weight. In *E. coli* it is about 2 nm thick and contains only one or two layers or sheets of peptidoglycan.

The outer membrane lies outside the thin peptidoglycan layer (figures 3.23 and 3.24). The most abundant membrane protein is Braun's lipoprotein, a small lipoprotein covalently joined to the underlying peptidoglycan and embedded in the outer membrane by its hydrophobic end. The outer membrane and peptidoglycan are so firmly linked by this lipoprotein that they can be isolated as one unit.

Another structure that may strengthen the gram-negative wall and hold the outer membrane in place is the adhesion site.

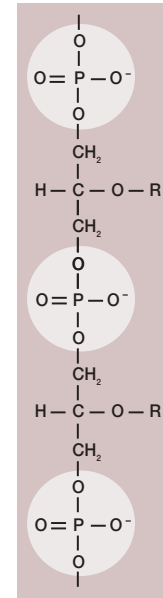


Figure 3.22 Teichoic Acid Structure. The segment of a teichoic acid made of phosphate, glycerol, and a side chain, R. R may represent D-alanine, glucose, or other molecules.

The outer membrane and plasma membrane appear to be in direct contact at many locations in the gram-negative wall. In *E. coli* 20 to 100 nm areas of contact between the two membranes are seen in plasmolyzed cells. Adhesion sites may be regions of direct contact or possibly true membrane fusions. It has been proposed that substances can move into the cell through these adhesion sites rather than traveling through the periplasm.

Possibly the most unusual constituents of the outer membrane are its **lipopolysaccharides (LPSs)**. These large, complex molecules contain both lipid and carbohydrate, and consist of three parts: (1) lipid A, (2) the core polysaccharide, and (3) the O side chain. The LPS from *Salmonella typhimurium* has been studied most, and its general structure is described here (figure 3.25). The **lipid A** region contains two glucosamine sugar derivatives, each with three fatty acids and phosphate or pyrophosphate attached. It is buried in the outer membrane and the remainder of the LPS molecule projects from the surface. The **core polysaccharide** is joined to lipid A. In *Salmonella* it is constructed of 10 sugars, many of them unusual in structure. The **O side chain** or **O antigen** is a polysaccharide chain extending outward from the core. It has several peculiar sugars and varies in composition between bacterial strains. Although O side chains are readily recognized by host antibodies, gram-negative bacteria may thwart host defenses by rapidly changing the nature of their O side chains to avoid detection. Antibody interaction with the LPS before reaching the outer membrane proper may also protect the cell wall from direct attack. **Antibodies and antigens** (chapters 32 and 33)

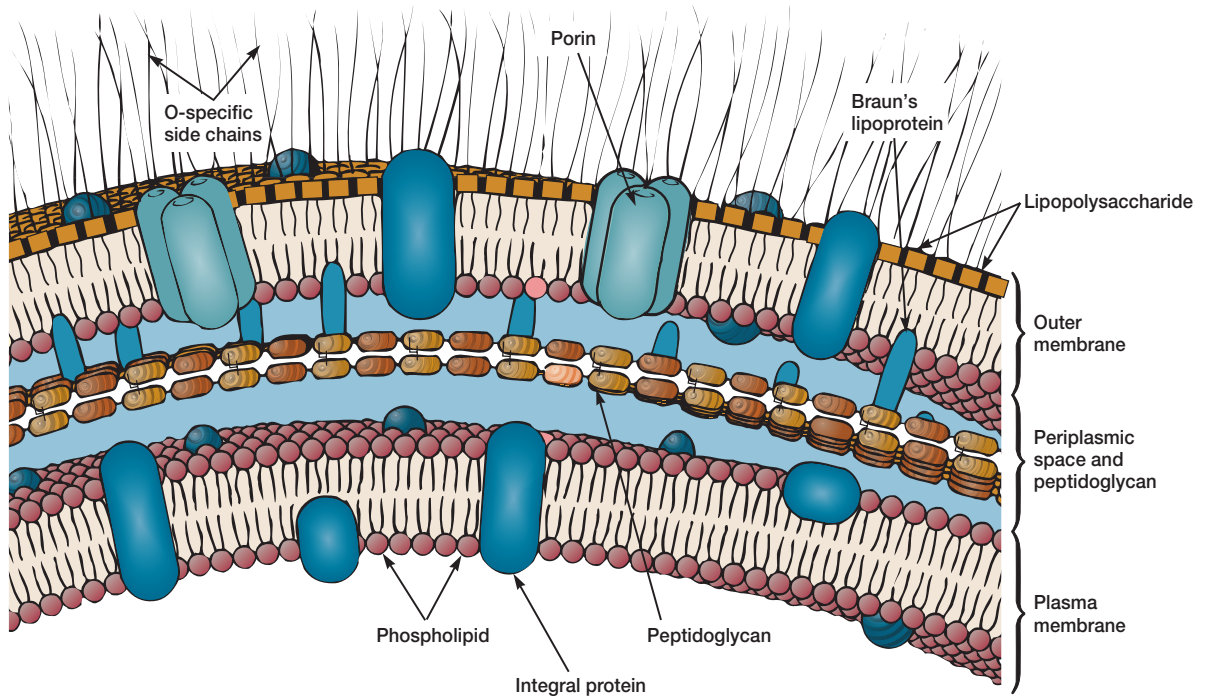


Figure 3.23 The Gram-Negative Envelope.

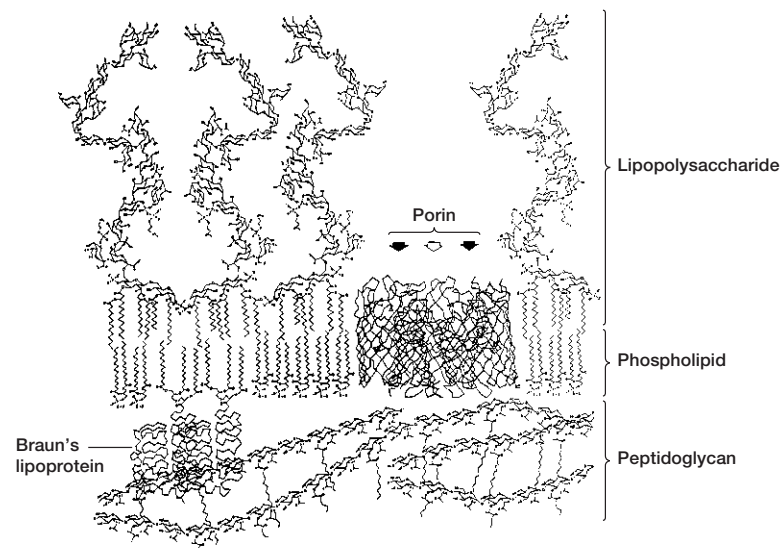
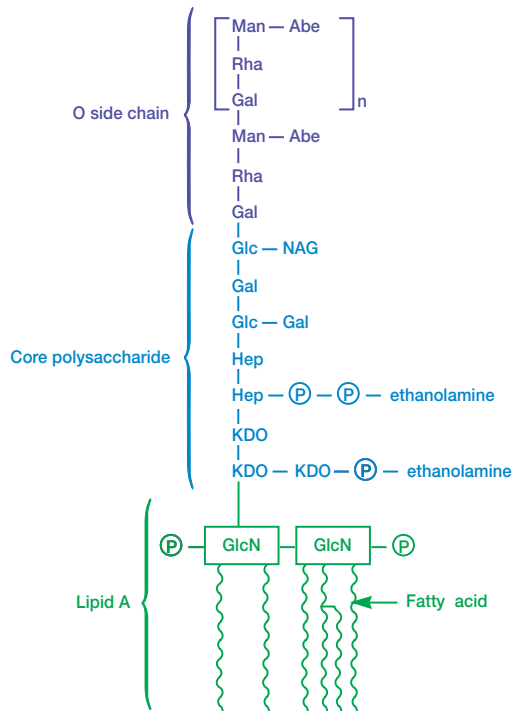
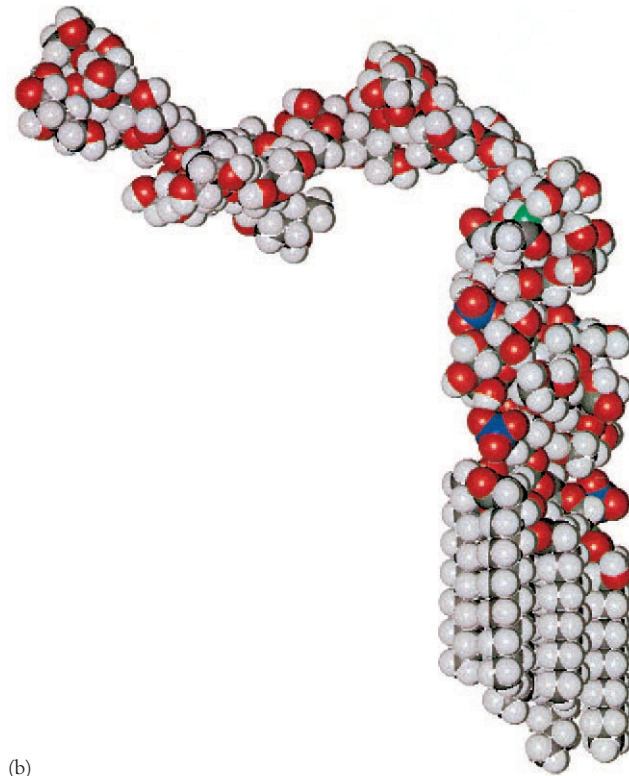


Figure 3.24 A Chemical Model of the *E. coli* Outer Membrane and Associated Structures. This cross-section is to-scale. The porin OmpF has two channels in the front (solid arrows) and one channel in the back (open arrow) of the trimeric protein complex. LPS molecules can be longer than the ones shown here.



(a)



(b)

Figure 3.25 Lipopolysaccharide Structure. (a) The lipopolysaccharide from *Salmonella*. This slightly simplified diagram illustrates one form of the LPS. Abbreviations: Abe, abequose; Gal, galactose; Glc, glucose; GlcN, glucosamine; Hep, heptulose; KDO, 2-keto-3-deoxyoctonate; Man, mannose; NAG, *N*-acetylglucosamine; P, phosphate; Rha, *L*-rhamnose. Lipid A is buried in the outer membrane. (b) Molecular model of an *Escherichia coli* lipopolysaccharide. The lipid A and core polysaccharide are straight; the O side chain is bent at an angle in this model.

The LPS is important for several reasons other than the avoidance of host defenses. Since the core polysaccharide usually contains charged sugars and phosphate (figure 3.25), LPS contributes to the negative charge on the bacterial surface. Lipid A is a major constituent of the outer membrane, and the LPS helps stabilize membrane structure. Furthermore, lipid A often is toxic; as a result the LPS can act as an endotoxin (*see section 34.3*) and cause some of the symptoms that arise in gram-negative bacterial infections.

A most important outer membrane function is to serve as a protective barrier. It prevents or slows the entry of bile salts, antibiotics, and other toxic substances that might kill or injure the bacterium. Even so, the outer membrane is more permeable than the plasma membrane and permits the passage of small molecules like glucose and other monosaccharides. This is due to the presence of special **porin proteins** (figures 3.23 and 3.24). Three porin molecules cluster together and span the outer membrane to form a narrow channel through which molecules smaller than about 600 to 700 daltons can pass. Larger molecules such as vitamin B₁₂ must be transported across the outer membrane by specific carriers. The outer membrane also prevents the loss of constituents like periplasmic enzymes.

The Mechanism of Gram Staining

Although several explanations have been given for the Gram-stain reaction results, it seems likely that the difference between gram-positive and gram-negative bacteria is due to the physical nature of their cell walls. If the cell wall is removed from gram-positive bacteria, they become gram negative. The peptidoglycan itself is not stained; instead it seems to act as a permeability barrier preventing loss of crystal violet. During the procedure the bacteria are first stained with crystal violet and next treated with iodine to promote dye retention. When gram-positive bacteria then are decolorized with ethanol, the alcohol is thought to shrink the pores of the thick peptidoglycan. Thus the dye-iodine complex is retained during the short decolorization step and the bacteria remain purple. In contrast, gram-negative peptidoglycan is very thin, not as highly cross-linked, and has larger pores. Alcohol treatment also may extract enough lipid from the gram-negative wall to increase its porosity further. For these reasons, alcohol more readily removes the purple crystal violet-iodine complex from gram-negative bacteria.

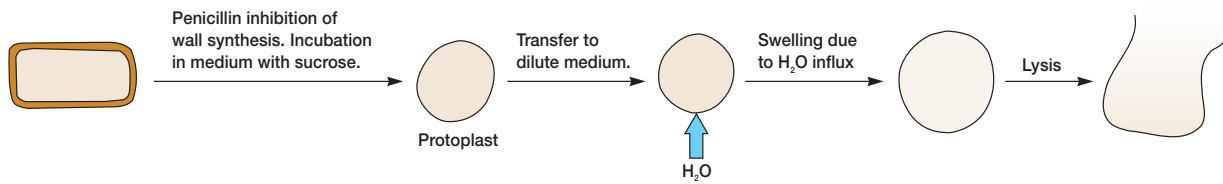


Figure 3.26 Protoplast Formation. Protoplast formation induced by incubation with penicillin in an isotonic medium. Transfer to dilute medium will result in lysis.

The Cell Wall and Osmotic Protection

The cell wall usually is required to protect bacteria against destruction by osmotic pressure. Solutes are much more concentrated in bacterial cytoplasm than in most microbial habitats, which are hypotonic. During **osmosis**, water moves across selectively permeable membranes such as the plasma membrane from dilute solutions (higher water concentration) to more concentrated solutions (lower water concentration). Thus water normally enters bacterial cells and the osmotic pressure may reach 20 atmospheres or 300 pounds/square inch. The plasma membrane cannot withstand such pressures and the cell will swell and be physically disrupted and destroyed, a process called **lysis**, without the wall that resists cell swelling and protects it. Solutes are more concentrated in hypertonic habitats than in the cell. Thus water flows outward, and the cytoplasm shrivels up and pulls away from the cell wall. This phenomenon is known as **plasmolysis** and is useful in food preservation because many microorganisms cannot grow in dried foods and jellies as they cannot avoid plasmolysis (see pp. 121–23, chapter 41).

The importance of the cell wall in protecting bacteria against osmotic lysis is demonstrated by treatment with lysozyme or penicillin. The enzyme **lysozyme** attacks peptidoglycan by hydrolyzing the bond that connects *N*-acetylmuramic acid with carbon four of *N*-acetylglucosamine. **Penicillin** inhibits peptidoglycan synthesis (see section 35.6). If bacteria are incubated with penicillin in an isotonic solution, gram-positive bacteria are converted to protoplasts that continue to grow normally when isotonicity is maintained even though they completely lack a wall. Gram-negative cells retain their outer membrane after penicillin treatment and are classified as **spheroplasts** because some of their cell wall remains. Protoplasts and spheroplasts are osmotically sensitive. If they are transferred to a dilute solution, they will lyse due to uncontrolled water influx (figure 3.26).

Although most bacteria require an intact cell wall for survival, some have none at all. For example, the mycoplasmas lack a cell wall and are osmotically sensitive, yet often can grow in dilute media or terrestrial environments because their plasma membranes are stronger than normal. The precise reason for this is not known, although the presence of sterols in the membranes of many species may provide added strength. Without a rigid cell wall, mycoplasmas tend to be pleomorphic or variable in shape.

1. Describe in some detail the composition and structure of peptidoglycan, gram-positive cell walls, and gram-negative cell walls. Include labeled diagrams in the answer.
2. Define or describe the following: outer membrane, periplasmic space, periplasm, envelope, teichoic acid, adhesion site, lipopolysaccharide, and porin protein.
3. Explain the role of the cell wall in protecting against lysis and how this role may be experimentally demonstrated. What are protoplasts and spheroplasts?

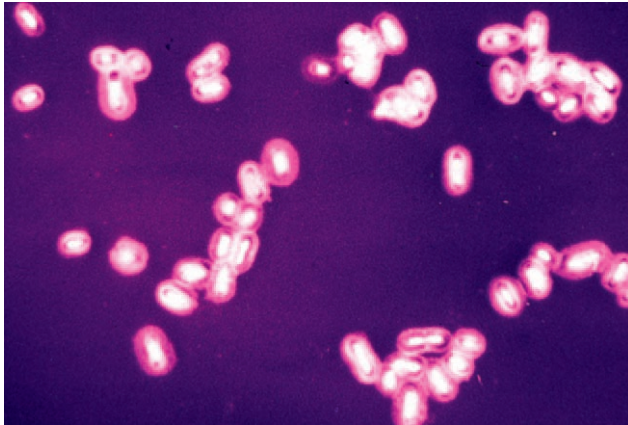
3.6 Components External to the Cell Wall

Bacteria have a variety of structures outside the cell wall that can function in protection, attachment to objects, and cell movement. Several of these are discussed.

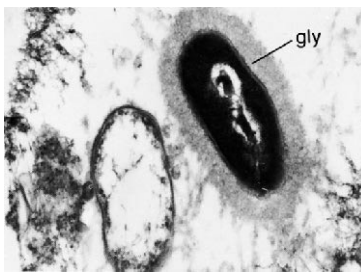
Capsules, Slime Layers, and S-Layers

Some bacteria have a layer of material lying outside the cell wall. When the layer is well organized and not easily washed off, it is called a **capsule**. A **slime layer** is a zone of diffuse, unorganized material that is removed easily. A **glycocalyx** (figure 3.27) is a network of polysaccharides extending from the surface of bacteria and other cells (in this sense it could encompass both capsules and slime layers). Capsules and slime layers usually are composed of polysaccharides, but they may be constructed of other materials. For example, *Bacillus anthracis* has a capsule of poly-D-glutamic acid. Capsules are clearly visible in the light microscope when negative stains or special capsule stains are employed (figure 3.27a); they also can be studied with the electron microscope (figure 3.27b).

Although capsules are not required for bacterial growth and reproduction in laboratory cultures, they do confer several advantages when bacteria grow in their normal habitats. They help bacteria resist phagocytosis by host phagocytic cells. *Streptococcus pneumoniae* provides a classic example. When it lacks a capsule, it is destroyed easily and does not cause disease, whereas the capsulated variant quickly kills mice. Capsules contain a great deal of water and can protect bacteria against desiccation. They



(a)



(b)

Figure 3.27 Bacterial Capsules. (a) *Klebsiella pneumoniae* with its capsule stained for observation in the light microscope ($\times 1,500$). (b) *Bacteroides* glycolyx (gly), TEM ($\times 71,250$).

exclude bacterial viruses and most hydrophobic toxic materials such as detergents. The glycolyx also aids bacterial attachment to surfaces of solid objects in aquatic environments or to tissue surfaces in plant and animal hosts (**figure 3.28**). Gliding bacteria often produce slime, which presumably aids in their motility (see *Box 21.1*). [The relationship of surface polysaccharides to phagocytosis and host colonization \(chapters 31 and 34\)](#)

Many gram-positive and gram-negative bacteria have a regularly structured layer called an **S-layer** on their surface. S-layers also are very common among Archaea, where they may be the only wall structure outside the plasma membrane. The S-layer has a pattern something like floor tiles and is composed of protein or glycoprotein (**figure 3.29**). In gram-negative bacteria the S-layer adheres directly to the outer membrane; it is associated with the peptidoglycan surface in gram-positive bacteria. It may protect the cell against ion and pH fluctuations, osmotic stress, enzymes, or the predacious bacterium *Bdellovibrio* (see *section 22.4*). The S-layer also helps maintain the shape and envelope rigidity of at least some bacterial cells. It can promote cell adhesion to surfaces. Finally, the layer seems to protect some pathogens against complement attack and phagocytosis, thus contributing to their virulence.

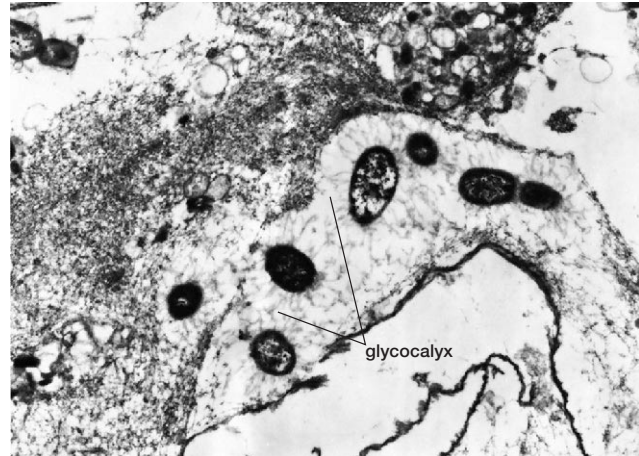


Figure 3.28 Bacterial Glycolyx. Bacteria connected to each other and to the intestinal wall, by their glycolyxes, the extensive networks of fibers extending from the cells ($\times 17,500$).

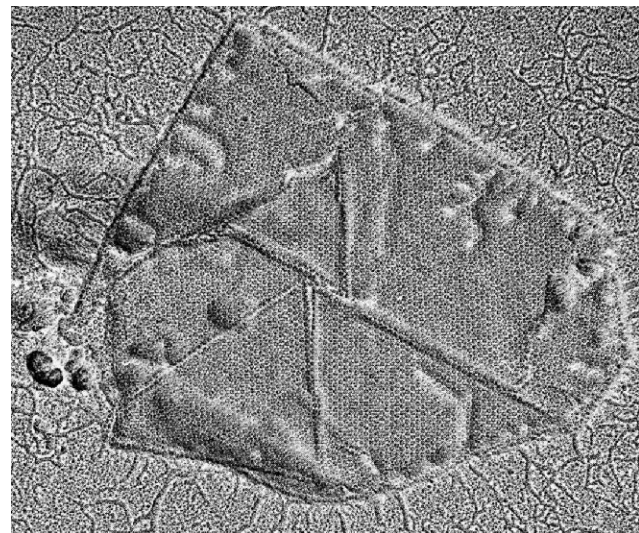


Figure 3.29 The S-Layer. An electron micrograph of the S-layer of *Deinococcus radiodurans* after shadowing.

Pili and Fimbriae

Many gram-negative bacteria have short, fine, hairlike appendages that are thinner than flagella and not involved in motility. These are usually called **fimbriae** (s., **fimbria**). Although a cell may be covered with up to 1,000 fimbriae, they are only visible in an electron microscope due to their small size (**figure 3.30**). They seem to be slender tubes composed of helically arranged protein subunits and are about 3 to 10 nm in diameter and up to several micrometers long. At least some types of fimbriae attach bacteria to solid surfaces such as rocks in streams and host tissues.

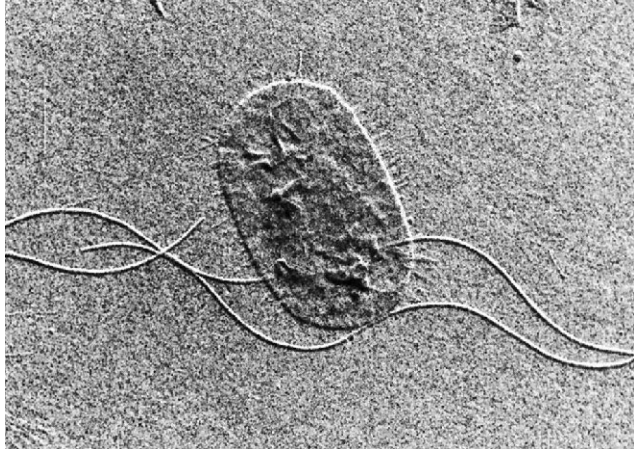


Figure 3.30 Flagella and Fimbriae. The long flagella and the numerous shorter fimbriae are very evident in this electron micrograph of *Proteus vulgaris* ($\times 39,000$).

Sex pili (s., **pilus**) are similar appendages, about 1 to 10 per cell, that differ from fimbriae in the following ways. Pili often are larger than fimbriae (around 9 to 10 nm in diameter). They are genetically determined by sex factors or conjugative plasmids and are required for bacterial mating (see chapter 13). Some bacterial viruses attach specifically to receptors on sex pili at the start of their reproductive cycle.

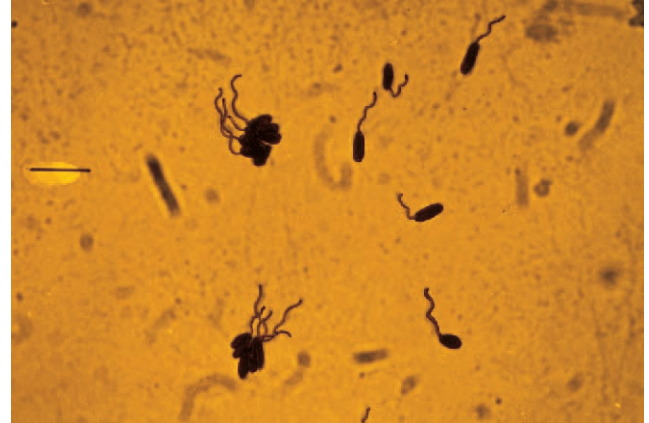
Flagella and Motility

Most motile bacteria move by use of **flagella** (s., **flagellum**), threadlike locomotor appendages extending outward from the plasma membrane and cell wall. They are slender, rigid structures, about 20 nm across and up to 15 or 20 μm long. Flagella are so thin they cannot be observed directly with a bright-field microscope, but must be stained with special techniques designed to increase their thickness (see chapter 2). The detailed structure of a flagellum can only be seen in the electron microscope (figure 3.30).

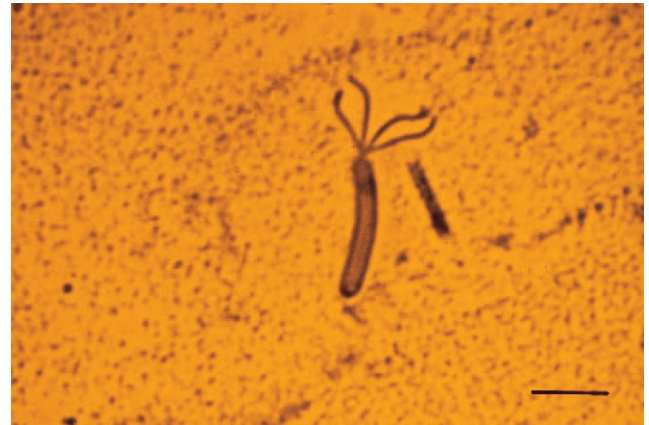
Bacterial species often differ distinctively in their patterns of flagella distribution. **Monotrichous** bacteria (*trichous* means hair) have one flagellum; if it is located at an end, it is said to be a **polar flagellum** (figure 3.31a). **Amphitrichous** bacteria (*amphi* means “on both sides”) have a single flagellum at each pole. In contrast, **lophotrichous** bacteria (*lopho* means tuft) have a cluster of flagella at one or both ends (figure 3.31b). Flagella are spread fairly evenly over the whole surface of **peritrichous** (*peri* means “around”) bacteria (figure 3.31c). Flagellation patterns are very useful in identifying bacteria.

Flagellar Ultrastructure

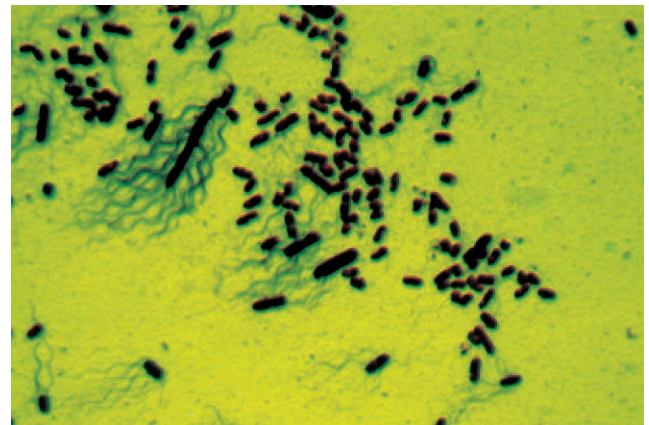
Transmission electron microscope studies have shown that the bacterial flagellum is composed of three parts. (1) The longest and most obvious portion is the **filament**, which extends from the



(a)



(b)



(c)

Figure 3.31 Flagellar Distribution. Examples of various patterns of flagellation as seen in the light microscope. (a) Monotrichous polar (*Pseudomonas*). (b) Lophotrichous (*Spirillum*). (c) Peritrichous (*Proteus vulgaris*, $\times 600$). Bars = 5 μm .

Figure 3.32 The Ultrastructure of Gram-Negative Flagella. (a) Negatively stained flagella from *Escherichia coli* ($\times 66,000$). Arrows indicate the location of curved hooks and basal bodies. (b) An enlarged view of the basal body of an *E. coli* flagellum ($\times 485,000$). All four rings (L, P, S, and M) can be clearly seen. The uppermost arrow is at the junction of the hook and filament. Bar = 30 nm.

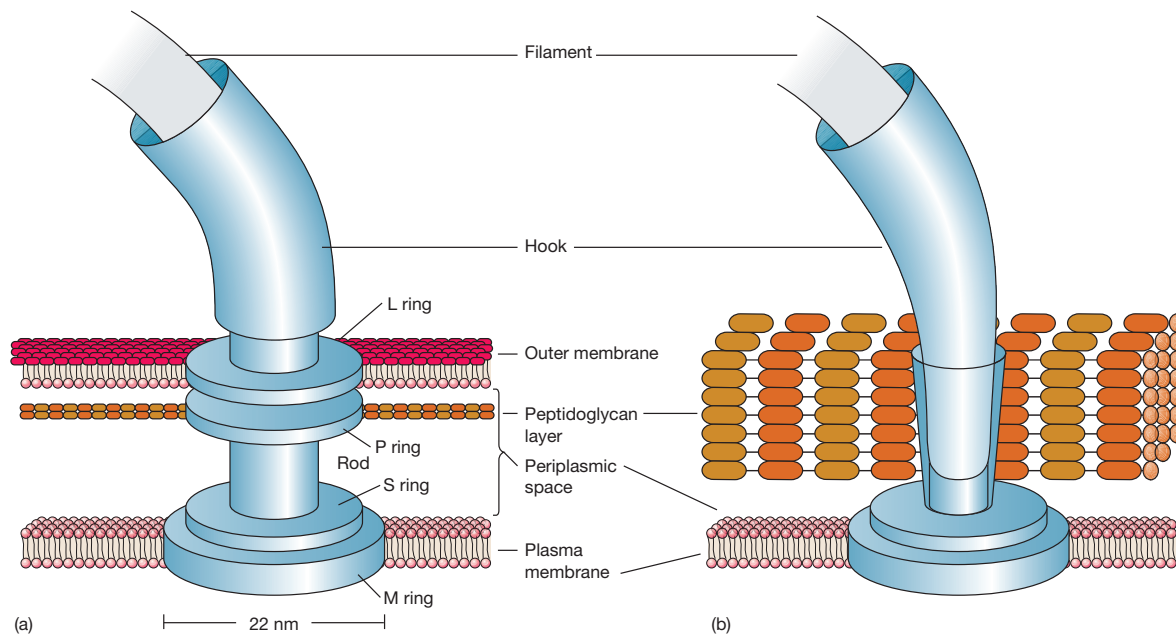
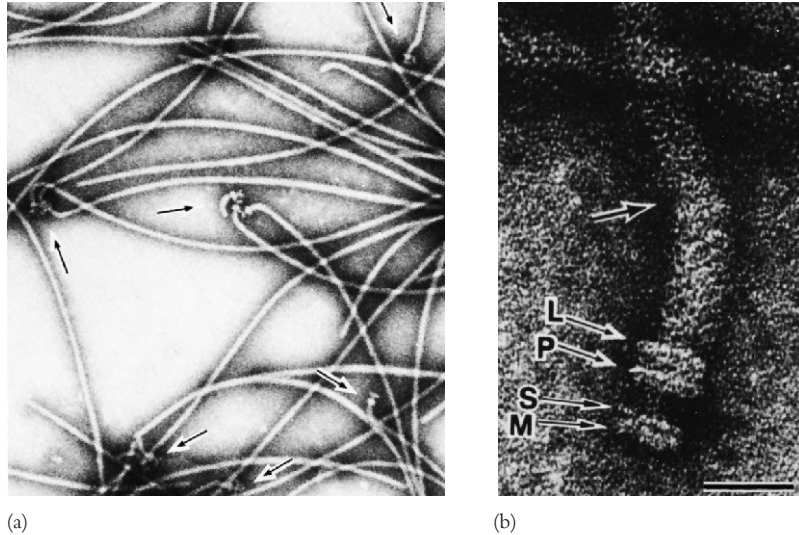


Figure 3.33 The Ultrastructure of Bacterial Flagella. Flagellar basal bodies and hooks in (a) gram-negative and (b) gram-positive bacteria.

cell surface to the tip. (2) A **basal body** is embedded in the cell; and (3) a short, curved segment, the **hook**, links the filament to its basal body and acts as a flexible coupling. The filament is a hollow, rigid cylinder constructed of a single protein called **flagellin**, which ranges in molecular weight from 30,000 to 60,000. The filament ends with a capping protein. Some bacteria have sheaths surrounding their flagella. For example *Bdellovibrio* has a membranous structure surrounding the filament. *Vibrio cholerae* has a lipopolysaccharide sheath.

The hook and basal body are quite different from the filament (**figure 3.32**). Slightly wider than the filament, the hook is made of different protein subunits. The basal body is the most complex part of a flagellum (**figure 3.32** and **figure 3.33**). In *E. coli* and most gram-negative bacteria, the body has four rings connected to a central rod. The outer L and P rings associate with the lipopolysaccharide and peptidoglycan layers, respectively. The inner M ring contacts the plasma membrane. Gram-positive bacteria have only two basal body rings, an inner ring

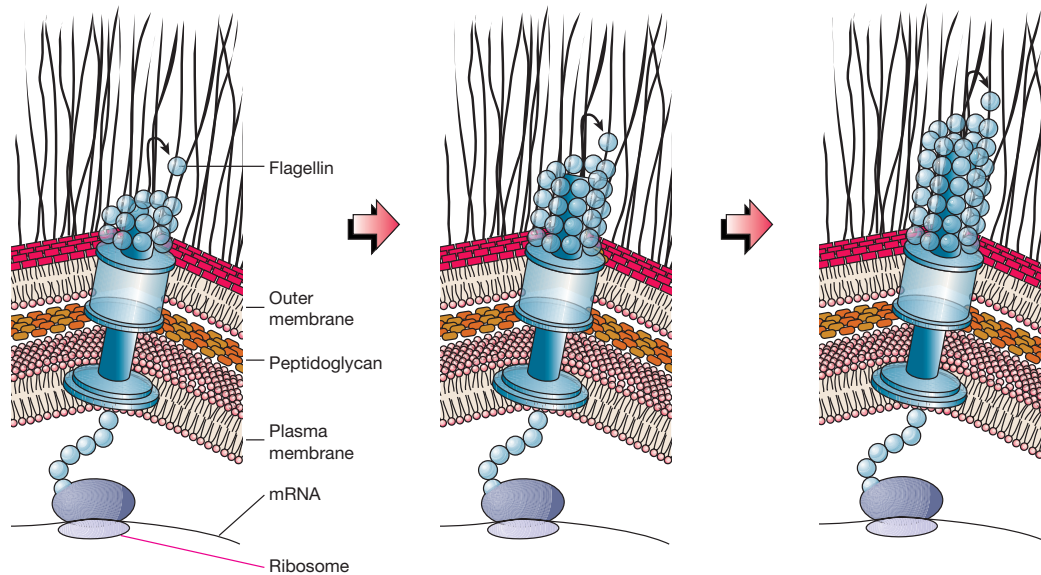


Figure 3.34 Growth of Flagellar Filaments. Flagellin subunits travel through the flagellar core and attach to the growing tip.

connected to the plasma membrane and an outer one probably attached to the peptidoglycan.

Flagellar Synthesis

The synthesis of flagella is a complex process involving at least 20 to 30 genes. Besides the gene for flagellin, 10 or more genes code for hook and basal body proteins; other genes are concerned with the control of flagellar construction or function. It is not known how the cell regulates or determines the exact location of flagella.

Bacteria can be deflagellated, and the regeneration of the flagellar filament can then be studied. It is believed that flagellin subunits are transported through the filament's hollow internal core. When they reach the tip, the subunits spontaneously aggregate under the direction of a special filament cap so that the filament grows at its tip rather than at the base (**figure 3.34**). Filament synthesis is an excellent example of **self-assembly**. Many structures form spontaneously through the association of their component parts without the aid of any special enzymes or other factors. The information required for filament construction is present in the structure of the flagellin subunit itself.

The Mechanism of Flagellar Movement

Prokaryotic flagella operate differently from eucaryotic flagella. The filament is in the shape of a rigid helix, and the bacterium moves when this helix rotates. Considerable evidence shows that flagella act just like propellers on a boat. Bacterial mutants with straight flagella or abnormally long hook regions (polyhook mutants) cannot swim. When bacteria are tethered to a glass slide us-

ing antibodies to filament or hook proteins, the cell body rotates rapidly about the stationary flagellum. If polystyrene-latex beads are attached to flagella, the beads spin about the flagellar axis due to flagellar rotation. The flagellar motor can rotate very rapidly. The *E. coli* motor rotates 270 revolutions per second; *Vibrio alginolyticus* averages 1,100 rps. [Eucaryotic flagella and motility \(pp. 89–90\)](#)

The direction of flagellar rotation determines the nature of bacterial movement. Monotrichous, polar flagella rotate counterclockwise (when viewed from outside the cell) during normal forward movement, whereas the cell itself rotates slowly clockwise. The rotating helical flagellar filament thrusts the cell forward in a run with the flagellum trailing behind (**figure 3.35**). Monotrichous bacteria stop and tumble randomly by reversing the direction of flagellar rotation. Peritrichously flagellated bacteria operate in a somewhat similar way. To move forward, the flagella rotate counterclockwise. As they do so, they bend at their hooks to form a rotating bundle that propels them forward. Clockwise rotation of the flagella disrupts the bundle and the cell tumbles.

Because bacteria swim through rotation of their rigid flagella, there must be some sort of motor at the base. A rod or shaft extends from the hook and ends in the M ring, which can rotate freely in the plasma membrane (**figure 3.36**). It is believed that the S ring is attached to the cell wall in gram-positive cells and does not rotate. The P and L rings of gram-negative bacteria would act as bearings for the rotating rod. There is some evidence that the basal body is a passive structure and rotates within a membrane-embedded protein complex much like the rotor of an electrical motor turns in the center of a ring of electromagnets (the stator).

The exact mechanism that drives basal body rotation still is not clear. Figure 3.36 provides a more detailed depiction of the basal